

Distribution Shift Results - Exact GP

Skin Lesion - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J3 - L8

Metric	Base						Scattering						ConvNeXt Atto						ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin			
LOSS	0.000 ± 0.000	0.650 ± 0.046	0.603 ± 0.041	0.627 ± 0.070	0.593 ± 0.062	0.619 ± 0.128	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002			
MAE	631.5 ± 21.9	361.2 ± 14.5	360.9 ± 14.0	355.8 ± 20.5	355.2 ± 21.9	473.8 ± 39.6	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5			
RMSE	814.9 ± 33.8	543.4 ± 28.3	541.8 ± 26.9	541.2 ± 31.6	538.3 ± 30.3	651.0 ± 37.0	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7			
NLL	1.517 ± 0.056	1.345 ± 0.189	1.372 ± 0.191	1.352 ± 0.223	1.398 ± 0.225	7.624 ± 7.928	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206			
QCE 95	-0.032 ± 0.020	-0.137 ± 0.020	-0.140 ± 0.019	-0.128 ± 0.024	-0.138 ± 0.025	-0.253 ± 0.113	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029			
PI 95 μ	3.920 ± 0.000	1.615 ± 0.075	1.582 ± 0.074	1.656 ± 0.075	1.615 ± 0.066	1.433 ± 0.174	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033			
PI 95 σ	0.000 ± 0.000	0.641 ± 0.052	0.631 ± 0.056	0.633 ± 0.036	0.643 ± 0.043	0.730 ± 0.062	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033			

J4 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.618 ± 0.045	0.603 ± 0.046	0.615 ± 0.042	0.555 ± 0.045	0.022 ± 0.002	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002
MAE	631.5 ± 21.9	348.0 ± 30.1	345.8 ± 31.9	351.3 ± 29.7	348.8 ± 32.9	319.1 ± 17.9	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5
RMSE	814.9 ± 33.8	538.9 ± 49.1	535.4 ± 49.7	545.5 ± 48.2	542.0 ± 49.2	511.6 ± 31.2	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7
NLL	1.517 ± 0.056	1.524 ± 0.237	1.594 ± 0.256	1.552 ± 0.240	1.692 ± 0.271	18.023 ± 2.715	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206
QCE 95	-0.032 ± 0.020	-0.140 ± 0.032	-0.142 ± 0.033	-0.140 ± 0.029	-0.147 ± 0.032	-0.553 ± 0.029	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029
PI 95 μ	3.920 ± 0.000	1.497 ± 0.062	1.449 ± 0.059	1.549 ± 0.063	1.489 ± 0.057	0.391 ± 0.046	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033
PI 95 σ	0.000 ± 0.000	0.593 ± 0.068	0.584 ± 0.066	0.595 ± 0.040	0.605 ± 0.034	0.195 ± 0.027	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033

J5 - L8

Metric	Base						Scattering						ConvNeXt Atto						ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin			
LOSS	0.000 ± 0.000	0.742 ± 0.033	0.681 ± 0.051	0.751 ± 0.029	0.634 ± 0.051	0.049 ± 0.005	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002			
MAE	631.5 ± 21.9	385.5 ± 18.1	384.4 ± 21.7	392.2 ± 18.3	396.6 ± 17.0	372.3 ± 28.0	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5			
RMSE	814.9 ± 33.8	584.4 ± 31.9	583.4 ± 34.5	592.6 ± 30.4	597.1 ± 24.5	575.0 ± 40.8	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7			
NLL	1.517 ± 0.056	1.783 ± 0.246	2.016 ± 0.432	1.808 ± 0.253	2.299 ± 0.410	27.956 ± 10.829	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206			
QCE 95	-0.032 ± 0.020	-0.150 ± 0.030	-0.157 ± 0.034	-0.148 ± 0.028	-0.176 ± 0.028	-0.624 ± 0.038	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029			
PI 95 μ	3.920 ± 0.000	1.634 ± 0.062	1.588 ± 0.052	1.716 ± 0.052	1.611 ± 0.044	0.387 ± 0.056	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033			
PI 95 σ	0.000 ± 0.000	0.583 ± 0.069	0.674 ± 0.073	0.632 ± 0.039	0.709 ± 0.036	0.176 ± 0.020	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033			

Skin Lesion - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J3 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.688 ± 0.036	0.682 ± 0.048	0.695 ± 0.039	0.697 ± 0.045	0.674 ± 0.107	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002
MAE	631.5 ± 21.9	377.9 ± 16.3	377.8 ± 16.3	380.5 ± 18.1	380.2 ± 18.2	496.3 ± 47.9	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5
RMSE	814.9 ± 33.8	579.5 ± 33.7	579.3 ± 33.7	589.0 ± 40.0	588.5 ± 39.9	673.8 ± 46.9	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7
NLL	1.517 ± 0.056	1.483 ± 0.217	1.487 ± 0.218	1.506 ± 0.292	1.508 ± 0.293	4.057 ± 2.431	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206
QCE 95	-0.032 ± 0.020	-0.149 ± 0.017	-0.148 ± 0.018	-0.148 ± 0.021	-0.148 ± 0.021	-0.274 ± 0.125	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029
PI 95 μ	3.920 ± 0.000	1.590 ± 0.065	1.585 ± 0.068	1.676 ± 0.063	1.672 ± 0.062	1.513 ± 0.166	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033
PI 95 σ	0.000 ± 0.000	0.545 ± 0.046	0.542 ± 0.049	0.628 ± 0.038	0.628 ± 0.039	0.708 ± 0.100	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033

J4 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.655 ± 0.055	0.646 ± 0.039	0.686 ± 0.035	0.680 ± 0.031	0.654 ± 0.144	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002
MAE	631.5 ± 21.9	369.2 ± 19.0	369.0 ± 19.3	378.5 ± 20.4	377.1 ± 20.8	479.0 ± 49.5	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5
RMSE	814.9 ± 33.8	576.9 ± 44.5	576.5 ± 44.5	578.5 ± 40.4	587.2 ± 50.4	653.7 ± 62.3	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7
NLL	1.517 ± 0.056	1.613 ± 0.251	1.621 ± 0.249	1.594 ± 0.282	1.604 ± 0.278	5.612 ± 4.066	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206
QCE 95	-0.032 ± 0.020	-0.158 ± 0.013	-0.158 ± 0.013	-0.159 ± 0.007	-0.157 ± 0.005	-0.268 ± 0.110	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029
PI 95 μ	3.920 ± 0.000	1.479 ± 0.064	1.473 ± 0.068	1.600 ± 0.057	1.588 ± 0.057	1.499 ± 0.206	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033
PI 95 σ	0.000 ± 0.000	0.468 ± 0.059	0.469 ± 0.069	0.597 ± 0.050	0.595 ± 0.047	0.828 ± 0.188	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033

J5 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.724 ± 0.022	0.708 ± 0.023	0.765 ± 0.024	0.757 ± 0.018	0.348 ± 0.098	0.780 ± 0.028	0.706 ± 0.043	0.788 ± 0.059	0.697 ± 0.050	0.070 ± 0.010	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002	0.947 ± 0.056	0.795 ± 0.051	0.948 ± 0.053	0.794 ± 0.053	0.014 ± 0.002
MAE	631.5 ± 21.9	402.6 ± 11.4	400.1 ± 12.4	410.3 ± 9.3	405.8 ± 10.9	400.8 ± 23.6	353.4 ± 22.7	365.3 ± 21.8	352.9 ± 22.2	367.7 ± 21.1	342.3 ± 27.8	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5	388.7 ± 18.1	394.3 ± 18.9	388.3 ± 18.6	391.1 ± 15.6	453.8 ± 19.5
RMSE	814.9 ± 33.8	609.5 ± 26.9	606.7 ± 28.6	623.6 ± 27.6	617.9 ± 29.1	592.8 ± 17.2	539.4 ± 38.8	552.1 ± 32.7	540.5 ± 39.6	555.6 ± 42.5	523.8 ± 38.6	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7	558.6 ± 19.8	563.8 ± 23.7	558.2 ± 21.0	561.0 ± 25.5	632.3 ± 26.7
NLL	1.517 ± 0.056	1.885 ± 0.202	1.896 ± 0.207	1.791 ± 0.195	1.816 ± 0.195	3.465 ± 1.062	1.649 ± 0.305	2.327 ± 0.560	1.620 ± 0.265	2.886 ± 0.828	37.805 ± 13.738	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206	1.250 ± 0.121	1.681 ± 0.598	1.239 ± 0.106	1.549 ± 0.295	44.824 ± 17.206
QCE 95	-0.032 ± 0.020	-0.170 ± 0.023	-0.173 ± 0.029	-0.163 ± 0.020	-0.164 ± 0.022	-0.192 ± 0.067	-0.135 ± 0.025	-0.180 ± 0.042	-0.136 ± 0.020	-0.193 ± 0.034	-0.627 ± 0.088	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029	-0.084 ± 0.028	-0.140 ± 0.057	-0.079 ± 0.022	-0.133 ± 0.038	-0.680 ± 0.029
PI 95 μ	3.920 ± 0.000	1.534 ± 0.065	1.517 ± 0.069	1.672 ± 0.050	1.637 ± 0.051	1.467 ± 0.063	1.609 ± 0.098	1.449 ± 0.108	1.643 ± 0.095	1.429 ± 0.109	0.347 ± 0.056	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033	2.220 ± 0.189	1.827 ± 0.236	2.241 ± 0.176	1.870 ± 0.162	0.443 ± 0.033
PI 95 σ	0.000 ± 0.000	0.408 ± 0.097	0.398 ± 0.105	0.562 ± 0.074	0.547 ± 0.068	0.750 ± 0.071	0.533 ± 0.053	0.582 ± 0.094	0.504 ± 0.064	0.586 ± 0.089	0.137 ± 0.012	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033	0.306 ± 0.033	0.383 ± 0.086	0.298 ± 0.021	0.353 ± 0.047	0.224 ± 0.033

Skin Lesion - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J3 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.050 ± 0.046	0.003 ± 0.041	0.027 ± 0.070	0.593 ± 0.062	0.619 ± 0.128	1.132 ± 0.038	1.029 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017
MAE	631.5 ± 21.9	361.2 ± 14.5	360.9 ± 14.0	355.8 ± 20.5	355.2 ± 21.9	473.8 ± 39.6	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7
RMSE	814.9 ± 33.8	543.4 ± 28.3	541.8 ± 26.9	541.2 ± 31.6	538.3 ± 30.3	651.0 ± 37.0	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2
NLL	1.517 ± 0.056	1.345 ± 0.189	1.372 ± 0.191	1.352 ± 0.223	1.398 ± 0.225	7.624 ± 7.928	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187
QCE 95	-0.032 ± 0.020	-0.137 ± 0.020	-0.140 ± 0.019	-0.128 ± 0.024	-0.138 ± 0.025	-0.253 ± 0.113	-0.151 ± 0.013	-0.156 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055
PI 95 μ	3.920 ± 0.000	1.615 ± 0.075	1.582 ± 0.074	1.656 ± 0.075	1.615 ± 0.066	1.433 ± 0.174	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073
PI 95 σ	0.000 ± 0.000	0.641 ± 0.052	0.631 ± 0.056	0.633 ± 0.036	0.643 ± 0.043	0.730 ± 0.062	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030

J4 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.618 ± 0.045	0.603 ± 0.046	0.615 ± 0.042	0.555 ± 0.045	0.022 ± 0.002	1.132 ± 0.038	1.029 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017
MAE	631.5 ± 21.9	348.0 ± 30.1	345.8 ± 31.9	351.3 ± 29.7	348.8 ± 32.9	319.1 ± 17.9	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7
RMSE	814.9 ± 33.8	538.9 ± 49.1	535.4 ± 49.7	545.5 ± 48.2	542.0 ± 49.2	511.6 ± 31.2	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2
NLL	1.517 ± 0.056	1.524 ± 0.237	1.594 ± 0.256	1.562 ± 0.240	1.692 ± 0.271	18.623 ± 2.715	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187
QCE 95	-0.032 ± 0.020	-0.140 ± 0.032	-0.142 ± 0.033	-0.140 ± 0.029	-0.147 ± 0.032	-0.553 ± 0.029	-0.151 ± 0.013	-0.150 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055
PI 95 μ	3.920 ± 0.000	1.497 ± 0.062	1.440 ± 0.059	1.540 ± 0.063	1.489 ± 0.057	0.391 ± 0.046	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073
PI 95 σ	0.000 ± 0.000	0.593 ± 0.068	0.584 ± 0.066	0.595 ± 0.040	0.605 ± 0.034	0.195 ± 0.027	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	0.742 ± 0.033	0.681 ± 0.051	0.751 ± 0.029	0.634 ± 0.051	0.049 ± 0.005	1.132 ± 0.038	1.029 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017
MAE	631.5 ± 21.9	305.5 ± 18.1	304.4 ± 21.7	302.2 ± 18.3	306.6 ± 17.0	372.3 ± 28.0	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7
RMSE	814.9 ± 33.8	584.4 ± 31.9	583.4 ± 34.5	592.6 ± 30.4	597.1 ± 24.5	575.0 ± 40.8	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2
NLL	1.517 ± 0.056	1.783 ± 0.246	2.016 ± 0.432	1.808 ± 0.253	2.299 ± 0.410	27.956 ± 10.829	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187
QCE 95	-0.032 ± 0.020	-0.150 ± 0.030	-0.157 ± 0.034	-0.148 ± 0.028	-0.176 ± 0.028	-0.624 ± 0.038	-0.151 ± 0.013	-0.156 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055
PI 95 μ	3.920 ± 0.000	1.634 ± 0.062	1.588 ± 0.052	1.716 ± 0.052	1.611 ± 0.044	0.387 ± 0.056	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073
PI 95 σ	0.000 ± 0.000	0.583 ± 0.069	0.674 ± 0.073	0.632 ± 0.039	0.709 ± 0.036	0.176 ± 0.020	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030

Skin Lesion - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J3 - L8

Metric	Base					Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	0.688 ± 0.036	0.682 ± 0.048	0.695 ± 0.039	0.697 ± 0.045	0.674 ± 0.107	1.132 ± 0.038	1.020 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017				
MAE	631.5 ± 21.9	377.9 ± 16.3	377.8 ± 16.3	380.5 ± 18.1	380.2 ± 18.2	496.3 ± 47.9	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7				
RMSE	814.9 ± 33.8	579.5 ± 33.7	579.3 ± 33.7	589.0 ± 40.0	588.5 ± 39.9	673.8 ± 46.9	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2				
NLL	1.517 ± 0.056	1.483 ± 0.217	1.487 ± 0.218	1.506 ± 0.292	1.508 ± 0.293	4.057 ± 2.431	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187				
QCE 95	-0.032 ± 0.020	-0.149 ± 0.017	-0.148 ± 0.018	-0.148 ± 0.021	-0.148 ± 0.021	-0.274 ± 0.125	-0.151 ± 0.013	-0.156 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055				
PI 95 μ	3.920 ± 0.000	1.590 ± 0.065	1.585 ± 0.068	1.676 ± 0.063	1.672 ± 0.062	1.513 ± 0.166	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073				
PI 95 σ	0.000 ± 0.000	0.545 ± 0.046	0.542 ± 0.049	0.628 ± 0.038	0.628 ± 0.039	0.708 ± 0.100	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030				

J4 - L8

Metric	Base						DINOv2 Small						DINOv2 Base					
	Base		Scattering				Base		Scattering				Base		Scattering			
	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	
LOSS	0.000 ± 0.000	0.655 ± 0.055	0.646 ± 0.039	0.686 ± 0.035	0.680 ± 0.031	0.654 ± 0.144	1.132 ± 0.038	1.020 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017		
MAE	631.5 ± 21.9	369.2 ± 19.0	369.0 ± 19.3	378.5 ± 20.4	377.1 ± 20.8	479.0 ± 49.5	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7		
RMSE	814.9 ± 33.8	576.9 ± 44.5	576.5 ± 44.5	588.9 ± 50.3	587.2 ± 50.4	653.7 ± 62.3	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2		
NLL	1.517 ± 0.056	1.613 ± 0.251	1.621 ± 0.249	1.594 ± 0.282	1.604 ± 0.278	5.612 ± 4.066	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187		
QCE 95	-0.032 ± 0.020	-0.150 ± 0.013	-0.150 ± 0.013	-0.150 ± 0.007	-0.157 ± 0.005	-0.248 ± 0.110	-0.151 ± 0.013	-0.156 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055		
PI 95 μ	3.920 ± 0.000	1.479 ± 0.064	1.473 ± 0.068	1.588 ± 0.057	1.588 ± 0.057	1.499 ± 0.206	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073		
PI 95 σ	0.000 ± 0.000	0.468 ± 0.059	0.469 ± 0.069	0.597 ± 0.050	0.595 ± 0.047	0.828 ± 0.188	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030		

J5 - L8

Metric	Base						DINOv2 Small						DINOv2 Base					
	RBF		RBF ARD		Matern		RBF		RBF ARD		Matern		RBF		RBF ARD		Matern	
LOSS	0.000 ± 0.000	0.724 ± 0.022	0.708 ± 0.023	0.765 ± 0.024	0.757 ± 0.018	0.348 ± 0.098	1.132 ± 0.038	1.020 ± 0.048	1.139 ± 0.033	1.013 ± 0.050	0.383 ± 0.029	1.051 ± 0.052	0.957 ± 0.057	1.055 ± 0.051	0.954 ± 0.054	0.186 ± 0.017		
MAE	631.5 ± 21.9	402.6 ± 11.4	400.1 ± 12.4	410.3 ± 9.3	405.8 ± 10.9	408.8 ± 23.6	565.0 ± 18.3	561.7 ± 40.0	568.3 ± 23.0	552.3 ± 37.6	543.1 ± 20.1	513.1 ± 17.3	520.9 ± 34.1	517.5 ± 25.2	530.1 ± 30.8	477.4 ± 30.7		
RMSE	814.9 ± 33.8	609.5 ± 26.9	606.7 ± 28.6	623.6 ± 27.6	617.9 ± 29.1	592.8 ± 17.2	782.2 ± 34.5	775.3 ± 43.3	783.4 ± 38.8	765.6 ± 40.7	757.9 ± 35.9	725.6 ± 26.0	730.3 ± 46.8	729.8 ± 33.8	739.2 ± 41.5	696.5 ± 48.2		
NLL	1.517 ± 0.056	1.885 ± 0.202	1.896 ± 0.207	1.791 ± 0.195	1.816 ± 0.195	3.465 ± 1.062	1.880 ± 0.137	2.115 ± 0.302	1.871 ± 0.150	2.038 ± 0.274	74.553 ± 95.320	1.815 ± 0.206	2.004 ± 0.286	1.804 ± 0.233	1.968 ± 0.294	62.010 ± 23.187		
QCE 95	-0.032 ± 0.020	-0.170 ± 0.023	-0.173 ± 0.029	-0.163 ± 0.020	-0.164 ± 0.022	-0.192 ± 0.027	-0.151 ± 0.013	-0.156 ± 0.020	-0.141 ± 0.013	-0.144 ± 0.013	-0.704 ± 0.096	-0.141 ± 0.015	-0.156 ± 0.032	-0.143 ± 0.023	-0.154 ± 0.022	-0.680 ± 0.055		
PI 95 μ	3.920 ± 0.000	1.534 ± 0.065	1.517 ± 0.069	1.672 ± 0.050	1.637 ± 0.051	1.467 ± 0.063	2.559 ± 0.081	2.664 ± 0.294	2.649 ± 0.094	2.683 ± 0.172	0.466 ± 0.170	2.491 ± 0.137	2.549 ± 0.204	2.585 ± 0.153	2.616 ± 0.165	0.388 ± 0.073		
PI 95 σ	0.000 ± 0.000	0.408 ± 0.097	0.398 ± 0.105	0.562 ± 0.074	0.547 ± 0.068	0.750 ± 0.071	1.200 ± 0.127	1.403 ± 0.082	1.136 ± 0.130	1.243 ± 0.066	0.134 ± 0.032	1.359 ± 0.054	1.429 ± 0.078	1.262 ± 0.085	1.256 ± 0.023	0.178 ± 0.030		

Histology Nuclei - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J3 - L8

Metric	Base						ConvNeXt Atto						ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		
LOSS	0.000 ± 0.000	1.126 ± 0.023	1.123 ± 0.020	1.116 ± 0.021	1.116 ± 0.025	0.557 ± 0.181	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014		
MAE	552.2 ± 20.5	309.6 ± 28.6	309.6 ± 28.7	308.1 ± 28.0	308.0 ± 28.2	390.6 ± 47.8	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2		
RMSE	635.3 ± 19.0	415.8 ± 43.3	415.7 ± 43.3	412.5 ± 42.7	412.2 ± 42.8	490.0 ± 37.0	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5		
NLL	1.326 ± 0.036	0.852 ± 0.067	0.853 ± 0.066	0.843 ± 0.067	0.845 ± 0.067	2.794 ± 1.537	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644		
QCE 95	0.041 ± 0.007	-0.004 ± 0.025	-0.005 ± 0.027	-0.004 ± 0.019	-0.004 ± 0.019	-0.182 ± 0.092	-0.027 ± 0.026	-0.016 ± 0.028	-0.018 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063		
PI 95 μ	3.920 ± 0.000	2.462 ± 0.067	2.469 ± 0.067	2.476 ± 0.073	2.473 ± 0.080	1.721 ± 0.143	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075		
PI 95 σ	0.000 ± 0.000	0.079 ± 0.008	0.079 ± 0.009	0.121 ± 0.014	0.122 ± 0.013	0.516 ± 0.065	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035		

J4 - L8

Metric	Base						ConvNeXt Atto						ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		
LOSS	0.000 ± 0.000	1.102 ± 0.028	1.101 ± 0.027	1.088 ± 0.020	1.088 ± 0.018	0.119 ± 0.019	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014		
MAE	552.2 ± 20.5	303.5 ± 29.2	303.4 ± 29.1	303.3 ± 29.2	303.2 ± 29.0	376.6 ± 29.5	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2		
RMSE	635.3 ± 19.0	407.9 ± 42.4	407.7 ± 42.0	408.8 ± 41.2	408.6 ± 41.0	521.5 ± 42.5	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5		
NLL	1.326 ± 0.036	0.828 ± 0.082	0.829 ± 0.082	0.828 ± 0.070	0.828 ± 0.071	13.613 ± 8.549	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644		
QCE 95	0.041 ± 0.007	-0.008 ± 0.024	-0.007 ± 0.024	-0.008 ± 0.021	-0.002 ± 0.020	-0.491 ± 0.075	-0.027 ± 0.026	-0.016 ± 0.028	-0.019 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063		
PI 95 μ	3.920 ± 0.000	2.276 ± 0.035	2.273 ± 0.038	2.433 ± 0.069	2.433 ± 0.069	0.655 ± 0.126	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075		
PI 95 σ	0.000 ± 0.000	0.070 ± 0.009	0.070 ± 0.008	0.114 ± 0.011	0.112 ± 0.012	0.178 ± 0.020	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035		

J5 - L8

Metric	Base						ConvNeXt Atto						ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		
LOSS	0.000 ± 0.000	1.098 ± 0.032	1.096 ± 0.036	1.072 ± 0.021	1.064 ± 0.027	0.138 ± 0.006	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014		
MAE	552.2 ± 20.5	305.4 ± 17.6	305.2 ± 17.5	308.2 ± 16.3	308.1 ± 16.1	321.6 ± 28.5	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2		
RMSE	635.3 ± 19.0	405.3 ± 27.6	405.2 ± 27.2	412.3 ± 26.5	411.8 ± 26.3	451.5 ± 36.9	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5		
NLL	1.326 ± 0.036	0.845 ± 0.083	0.845 ± 0.083	0.840 ± 0.062	0.840 ± 0.061	16.418 ± 8.052	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644		
QCE 95	0.041 ± 0.007	-0.026 ± 0.017	-0.027 ± 0.017	-0.010 ± 0.018	-0.008 ± 0.018	-0.538 ± 0.067	-0.027 ± 0.026	-0.016 ± 0.028	-0.018 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063		
PI 95 μ	3.920 ± 0.000	2.062 ± 0.027	2.063 ± 0.030	2.304 ± 0.052	2.303 ± 0.069	0.504 ± 0.089	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075		
PI 95 σ	0.000 ± 0.000	0.073 ± 0.005	0.072 ± 0.005	0.128 ± 0.007	0.126 ± 0.007	0.145 ± 0.018	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035		

Histology Nuclei - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J3 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.129 ± 0.029	1.129 ± 0.027	1.119 ± 0.025	1.117 ± 0.025	0.907 ± 0.144	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014
MAE	552.2 ± 20.5	303.6 ± 28.2	303.6 ± 28.2	300.2 ± 27.7	300.1 ± 27.6	516.8 ± 37.8	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2
RMSE	635.3 ± 19.0	402.9 ± 44.6	402.9 ± 44.6	395.3 ± 42.4	395.2 ± 42.4	601.5 ± 33.7	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5
NLL	1.326 ± 0.036	0.850 ± 0.070	0.850 ± 0.069	0.840 ± 0.068	0.840 ± 0.067	51993984.547 ± 107785214.184	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644
QCE 95	0.041 ± 0.007	0.002 ± 0.027	0.001 ± 0.027	0.009 ± 0.020	0.009 ± 0.020	-0.728 ± 0.755	-0.027 ± 0.026	-0.016 ± 0.028	-0.018 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063
PI 95 μ	3.920 ± 0.000	2.480 ± 0.067	2.474 ± 0.064	2.478 ± 0.079	2.485 ± 0.077	0.645 ± 0.647	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075
PI 95 σ	0.000 ± 0.000	0.069 ± 0.009	0.069 ± 0.009	0.118 ± 0.015	0.117 ± 0.015	0.155 ± 0.206	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035

J4 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.116 ± 0.021	1.114 ± 0.029	1.099 ± 0.027	1.099 ± 0.028	0.822 ± 0.044	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014
MAE	552.2 ± 20.5	299.4 ± 24.6	299.5 ± 24.7	298.0 ± 29.2	298.0 ± 29.0	484.9 ± 9.1	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2
RMSE	635.3 ± 19.0	391.9 ± 42.8	391.9 ± 42.9	391.9 ± 45.8	391.8 ± 45.5	567.8 ± 6.1	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5
NLL	1.326 ± 0.036	0.821 ± 0.075	0.821 ± 0.075	0.821 ± 0.068	0.820 ± 0.069	4.835 ± 1.906	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644
QCE 95	0.041 ± 0.007	0.004 ± 0.027	0.004 ± 0.027	0.012 ± 0.021	0.012 ± 0.021	-0.464 ± 0.101	-0.027 ± 0.026	-0.016 ± 0.028	-0.018 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063
PI 95 μ	3.920 ± 0.000	2.338 ± 0.047	2.336 ± 0.047	2.464 ± 0.069	2.464 ± 0.074	1.277 ± 0.250	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075
PI 95 σ	0.000 ± 0.000	0.052 ± 0.007	0.052 ± 0.007	0.094 ± 0.012	0.094 ± 0.012	0.245 ± 0.135	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035

J5 - L8

Metric	Base	Scattering					ConvNeXt Atto					ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.117 ± 0.029	1.113 ± 0.033	1.083 ± 0.023	1.077 ± 0.027	0.335 ± 0.059	1.112 ± 0.029	1.077 ± 0.032	1.084 ± 0.032	1.058 ± 0.024	0.204 ± 0.019	1.209 ± 0.038	1.077 ± 0.030	1.136 ± 0.042	1.005 ± 0.028	0.071 ± 0.014
MAE	552.2 ± 20.5	304.6 ± 14.8	304.6 ± 14.8	304.2 ± 18.1	304.4 ± 18.2	345.3 ± 4.2	382.6 ± 41.5	388.0 ± 42.4	366.9 ± 24.8	364.5 ± 31.1	365.2 ± 37.0	381.0 ± 19.5	379.9 ± 18.6	367.0 ± 11.2	372.1 ± 13.8	421.0 ± 46.2
RMSE	635.3 ± 19.0	396.4 ± 26.0	396.5 ± 26.9	396.1 ± 30.9	396.3 ± 31.1	456.4 ± 24.4	510.3 ± 47.5	515.5 ± 47.0	497.3 ± 39.4	490.2 ± 49.7	498.9 ± 57.5	519.0 ± 32.3	511.4 ± 27.8	512.7 ± 26.3	506.0 ± 16.7	568.5 ± 42.5
NLL	1.326 ± 0.036	0.829 ± 0.060	0.829 ± 0.069	0.825 ± 0.057	0.825 ± 0.058	2.931 ± 0.582	0.892 ± 0.091	0.890 ± 0.082	0.881 ± 0.090	0.876 ± 0.086	14.531 ± 8.579	1.169 ± 0.206	1.228 ± 0.223	0.961 ± 0.091	1.092 ± 0.193	65.692 ± 39.644
QCE 95	0.041 ± 0.007	-0.008 ± 0.012	-0.009 ± 0.011	0.007 ± 0.019	0.006 ± 0.020	-0.200 ± 0.024	-0.027 ± 0.026	-0.016 ± 0.028	-0.018 ± 0.023	-0.010 ± 0.024	-0.536 ± 0.084	-0.101 ± 0.025	-0.107 ± 0.023	-0.067 ± 0.015	-0.102 ± 0.028	-0.703 ± 0.063
PI 95 μ	3.920 ± 0.000	2.189 ± 0.025	2.183 ± 0.021	2.387 ± 0.044	2.381 ± 0.055	1.392 ± 0.066	2.728 ± 0.086	2.849 ± 0.116	2.835 ± 0.094	2.910 ± 0.150	0.604 ± 0.113	2.476 ± 0.127	2.332 ± 0.115	2.543 ± 0.129	2.174 ± 0.082	0.398 ± 0.075
PI 95 σ	0.000 ± 0.000	0.042 ± 0.003	0.042 ± 0.003	0.082 ± 0.004	0.082 ± 0.005	0.418 ± 0.049	0.515 ± 0.055	0.545 ± 0.051	0.471 ± 0.045	0.486 ± 0.040	0.182 ± 0.052	1.102 ± 0.053	0.997 ± 0.049	0.712 ± 0.187	0.679 ± 0.039	0.203 ± 0.035

Histogram Nuclei - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J3 - L8

Metric	Base					DINOv2 Small					DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.126 ± 0.023	1.123 ± 0.020	1.116 ± 0.021	1.116 ± 0.025	0.557 ± 0.181	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010
MAE	552.2 ± 20.5	309.6 ± 28.6	309.6 ± 28.7	308.1 ± 28.0	308.0 ± 28.2	390.6 ± 47.8	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7
RMSE	635.3 ± 19.0	415.8 ± 43.3	415.7 ± 43.3	412.5 ± 42.7	412.2 ± 42.8	490.0 ± 37.0	452.2 ± 50.2	485.8 ± 39.4	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9
NLL	1.326 ± 0.036	0.852 ± 0.067	0.853 ± 0.066	0.843 ± 0.067	0.845 ± 0.067	2.794 ± 1.537	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953
QCE 95	0.041 ± 0.007	-0.004 ± 0.025	-0.005 ± 0.027	-0.004 ± 0.019	-0.004 ± 0.019	-0.182 ± 0.092	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032
PI 95 μ	3.920 ± 0.000	2.462 ± 0.067	2.469 ± 0.067	2.476 ± 0.073	2.473 ± 0.080	1.721 ± 0.143	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035
PI 95 σ	0.000 ± 0.000	0.079 ± 0.008	0.079 ± 0.009	0.121 ± 0.014	0.122 ± 0.013	0.516 ± 0.065	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028

J4 - L8

Metric	Base					ScatterNet					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	1.102 ± 0.028	1.101 ± 0.027	1.088 ± 0.020	1.088 ± 0.019	0.119 ± 0.019	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010				
MAE	552.2 ± 20.5	303.5 ± 29.2	303.4 ± 29.1	303.3 ± 29.2	303.2 ± 29.0	376.6 ± 29.5	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7				
RMSE	635.3 ± 19.0	407.9 ± 42.2	407.7 ± 42.0	408.8 ± 41.2	408.6 ± 41.0	521.5 ± 42.5	452.2 ± 50.2	485.8 ± 39.4	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9				
NLL	1.326 ± 0.036	0.828 ± 0.082	0.829 ± 0.082	0.828 ± 0.070	0.828 ± 0.071	13.613 ± 8.549	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953				
QCE 95	0.041 ± 0.007	-0.008 ± 0.024	-0.007 ± 0.024	-0.009 ± 0.021	-0.002 ± 0.020	-0.491 ± 0.075	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032				
PI 95 μ	3.920 ± 0.000	2.276 ± 0.035	2.273 ± 0.038	2.433 ± 0.069	2.433 ± 0.069	0.655 ± 0.126	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035				
PI 95 σ	0.000 ± 0.000	0.070 ± 0.009	0.070 ± 0.008	0.114 ± 0.011	0.112 ± 0.012	0.178 ± 0.020	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028				

J5 - L8

Metric	Base					Scatterer					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin				
LOSS	0.000 ± 0.000	1.098 ± 0.032	1.096 ± 0.036	1.072 ± 0.021	1.064 ± 0.027	0.138 ± 0.006	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010				
MAE	552.2 ± 20.5	305.4 ± 17.6	305.2 ± 17.5	308.2 ± 16.3	308.1 ± 16.1	321.6 ± 28.5	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7				
RMSE	635.3 ± 19.0	405.7 ± 27.6	405.2 ± 27.2	412.3 ± 26.5	411.8 ± 26.3	451.5 ± 36.9	452.2 ± 50.2	485.8 ± 39.4	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9				
NLL	1.326 ± 0.036	0.845 ± 0.083	0.845 ± 0.083	0.840 ± 0.062	0.840 ± 0.061	16.418 ± 8.052	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953				
QCE 95	0.041 ± 0.007	-0.026 ± 0.017	-0.027 ± 0.017	-0.010 ± 0.018	-0.008 ± 0.018	-0.538 ± 0.067	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032				
PI 95 μ	3.920 ± 0.000	2.062 ± 0.027	2.063 ± 0.030	2.304 ± 0.052	2.303 ± 0.069	0.504 ± 0.089	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035				
PI 95 σ	0.000 ± 0.000	0.073 ± 0.005	0.072 ± 0.005	0.128 ± 0.007	0.126 ± 0.007	0.145 ± 0.018	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028				

Histology Nuclei - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J3 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.129 ± 0.029	1.129 ± 0.027	1.119 ± 0.025	1.117 ± 0.025	0.907 ± 0.144	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010
MAE	552.2 ± 20.5	303.6 ± 28.2	303.6 ± 28.2	300.2 ± 27.7	300.1 ± 27.6	516.6 ± 37.8	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7
RMSE	635.3 ± 19.0	402.9 ± 44.6	402.9 ± 44.6	395.3 ± 42.4	395.2 ± 42.4	601.5 ± 33.7	452.2 ± 50.2	485.8 ± 30.4	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9
NLL	1.326 ± 0.036	0.850 ± 0.070	0.850 ± 0.069	0.840 ± 0.068	0.840 ± 0.067	51993984.547 ± 107785214.184	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953
QCE 95	0.041 ± 0.007	0.002 ± 0.027	0.001 ± 0.027	0.009 ± 0.020	0.009 ± 0.020	-0.728 ± 0.255	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032
PI 95 μ	3.920 ± 0.000	2.480 ± 0.067	2.474 ± 0.064	2.478 ± 0.079	2.485 ± 0.077	0.445 ± 0.647	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035
PI 95 σ	0.000 ± 0.000	0.069 ± 0.009	0.069 ± 0.009	0.118 ± 0.015	0.117 ± 0.015	0.155 ± 0.206	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028

J4 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.116 ± 0.021	1.114 ± 0.029	1.099 ± 0.027	1.099 ± 0.028	0.822 ± 0.044	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010
MAE	552.2 ± 20.5	299.4 ± 24.6	299.5 ± 24.7	298.0 ± 29.2	298.0 ± 29.0	484.9 ± 9.1	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7
RMSE	635.3 ± 19.0	391.9 ± 42.8	391.9 ± 42.9	391.9 ± 45.8	391.8 ± 45.5	567.8 ± 6.1	452.2 ± 50.2	485.8 ± 30.8	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9
NLL	1.326 ± 0.036	0.821 ± 0.075	0.821 ± 0.075	0.821 ± 0.068	0.820 ± 0.069	4.835 ± 1.906	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953
QCE 95	0.041 ± 0.007	0.004 ± 0.027	0.004 ± 0.027	0.012 ± 0.021	0.012 ± 0.021	-0.464 ± 0.101	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032
PI 95 μ	3.920 ± 0.000	2.338 ± 0.047	2.336 ± 0.047	2.464 ± 0.069	2.464 ± 0.074	1.277 ± 0.250	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035
PI 95 σ	0.000 ± 0.000	0.052 ± 0.007	0.052 ± 0.007	0.094 ± 0.012	0.094 ± 0.012	0.245 ± 0.335	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.117 ± 0.029	1.113 ± 0.033	1.083 ± 0.023	1.077 ± 0.027	0.335 ± 0.059	1.094 ± 0.037	1.094 ± 0.042	1.074 ± 0.043	1.045 ± 0.062	0.340 ± 0.022	1.045 ± 0.026	1.033 ± 0.027	1.011 ± 0.034	1.006 ± 0.035	0.228 ± 0.010
MAE	552.2 ± 20.5	304.6 ± 14.8	304.6 ± 14.8	304.2 ± 18.1	304.4 ± 18.2	345.3 ± 4.2	328.5 ± 35.9	367.8 ± 30.8	326.6 ± 34.5	341.7 ± 31.8	305.3 ± 24.8	325.7 ± 10.8	325.6 ± 10.4	330.2 ± 17.2	329.4 ± 16.9	306.4 ± 18.7
RMSE	635.3 ± 19.0	396.3 ± 26.0	396.5 ± 26.0	390.1 ± 30.9	390.3 ± 31.1	456.4 ± 24.4	452.2 ± 50.2	485.8 ± 30.8	446.9 ± 48.2	459.7 ± 46.5	415.0 ± 35.6	436.6 ± 18.1	437.2 ± 16.5	447.6 ± 23.8	446.9 ± 24.0	410.4 ± 24.9
NLL	1.326 ± 0.036	0.825 ± 0.068	0.825 ± 0.069	0.825 ± 0.057	0.825 ± 0.058	2.031 ± 0.582	0.902 ± 0.148	0.979 ± 0.150	0.892 ± 0.137	0.903 ± 0.105	46.438 ± 28.388	0.855 ± 0.120	0.844 ± 0.110	0.841 ± 0.103	0.835 ± 0.101	25.285 ± 7.953
QCE 95	0.041 ± 0.007	-0.008 ± 0.012	-0.009 ± 0.011	0.007 ± 0.019	0.006 ± 0.020	-0.200 ± 0.024	-0.088 ± 0.027	-0.072 ± 0.030	-0.068 ± 0.026	-0.048 ± 0.028	-0.693 ± 0.064	-0.075 ± 0.022	-0.071 ± 0.020	-0.071 ± 0.025	-0.066 ± 0.022	-0.627 ± 0.032
PI 95 μ	3.920 ± 0.000	2.189 ± 0.025	2.183 ± 0.021	2.387 ± 0.044	2.381 ± 0.055	1.392 ± 0.066	2.076 ± 0.052	2.579 ± 0.055	2.195 ± 0.031	2.552 ± 0.086	0.306 ± 0.126	2.025 ± 0.081	2.062 ± 0.076	2.170 ± 0.096	2.192 ± 0.094	0.405 ± 0.035
PI 95 σ	0.000 ± 0.000	0.042 ± 0.003	0.042 ± 0.003	0.082 ± 0.004	0.082 ± 0.005	0.418 ± 0.049	0.758 ± 0.072	1.217 ± 0.143	0.826 ± 0.021	1.059 ± 0.091	0.099 ± 0.021	0.471 ± 0.112	0.487 ± 0.112	0.548 ± 0.103	0.553 ± 0.106	0.157 ± 0.028

Poverty - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs ConvNeXt

J5 - L8

Metric	Base		Scattering				ConvNeXt Atto				ConvNeXt Base					
	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	
LOSS	0.000 ± 0.000	1.061 ± 0.010	0.525 ± 0.024	1.045 ± 0.010	0.547 ± 0.040	0.029 ± 0.010	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.552 ± 0.019	0.544 ± 0.050	0.528 ± 0.020	0.508 ± 0.051	0.558 ± 0.039	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.676 ± 0.022	0.700 ± 0.070	0.648 ± 0.025	0.653 ± 0.074	0.714 ± 0.059	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.040	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.101 ± 0.056	3.897 ± 0.453	1.064 ± 0.045	3.054 ± 0.438	41.557 ± 25.712	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.038 ± 0.013	-0.363 ± 0.049	-0.038 ± 0.014	-0.335 ± 0.048	-0.753 ± 0.041	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	2.881 ± 0.048	1.844 ± 0.118	2.801 ± 0.023	1.879 ± 0.110	0.505 ± 0.123	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	0.818 ± 0.101	1.440 ± 0.135	0.676 ± 0.036	1.419 ± 0.075	0.282 ± 0.034	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

J6 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.029 ± 0.012	0.707 ± 0.038	1.016 ± 0.019	0.697 ± 0.038	0.104 ± 0.004	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.532 ± 0.023	0.517 ± 0.034	0.518 ± 0.018	0.488 ± 0.033	0.564 ± 0.044	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.656 ± 0.031	0.657 ± 0.050	0.639 ± 0.026	0.612 ± 0.044	0.748 ± 0.081	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.040	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.070 ± 0.051	2.477 ± 0.496	1.059 ± 0.049	2.114 ± 0.259	214.390 ± 121.202	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.032 ± 0.015	-0.247 ± 0.022	-0.031 ± 0.014	-0.236 ± 0.043	-0.864 ± 0.026	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	2.743 ± 0.030	2.213 ± 0.153	2.734 ± 0.029	2.189 ± 0.141	0.223 ± 0.063	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	0.664 ± 0.032	1.456 ± 0.080	0.669 ± 0.038	1.397 ± 0.062	0.196 ± 0.101	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

J7 - L8

Metric	Base					Scattering					ConvNeXt Atto					ConvNeXt Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.046 ± 0.025	0.883 ± 0.024	1.035 ± 0.021	0.850 ± 0.036	0.281 ± 0.009	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.532 ± 0.019	0.535 ± 0.014	0.519 ± 0.023	0.490 ± 0.010	0.529 ± 0.033	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.659 ± 0.029	0.666 ± 0.007	0.642 ± 0.022	0.611 ± 0.009	0.701 ± 0.087	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.040	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.111 ± 0.057	1.479 ± 0.181	1.086 ± 0.089	1.323 ± 0.122	1327.534 ± 819.459	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.035 ± 0.012	-0.148 ± 0.042	-0.029 ± 0.010	-0.128 ± 0.043	-0.908 ± 0.010	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	2.826 ± 0.149	2.438 ± 0.150	2.770 ± 0.089	2.343 ± 0.160	0.089 ± 0.020	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	0.726 ± 0.219	1.164 ± 0.084	0.655 ± 0.112	1.062 ± 0.065	0.097 ± 0.060	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

Poverty - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs ConvNeXt

J5 - L8

Metric	Base						ConvNeXt Atto										ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.077 ± 0.016	0.501 ± 0.042	1.059 ± 0.013	0.401 ± 0.045	0.008 ± 0.003	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.551 ± 0.032	0.500 ± 0.049	0.530 ± 0.025	0.513 ± 0.042	0.511 ± 0.033	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.690 ± 0.043	0.630 ± 0.051	0.658 ± 0.032	0.653 ± 0.048	0.646 ± 0.044	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.040	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.125 ± 0.043	3.021 ± 0.639	1.111 ± 0.037	3.367 ± 0.387	34.393 ± 10.812	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.058 ± 0.016	-0.326 ± 0.044	-0.045 ± 0.020	-0.370 ± 0.042	-0.748 ± 0.026	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	2.930 ± 0.076	1.745 ± 0.146	2.950 ± 0.073	1.589 ± 0.056	0.419 ± 0.069	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	1.000 ± 0.161	1.112 ± 0.056	1.079 ± 0.143	1.021 ± 0.055	0.139 ± 0.014	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

J6 - L8

Metric	Base						ConvNeXt Atto										ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.057 ± 0.003	0.465 ± 0.015	1.052 ± 0.006	0.476 ± 0.029	0.013 ± 0.005	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.550 ± 0.027	0.530 ± 0.056	0.531 ± 0.023	0.511 ± 0.044	0.521 ± 0.039	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.687 ± 0.041	0.670 ± 0.064	0.660 ± 0.033	0.646 ± 0.051	0.657 ± 0.050	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.043	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.140 ± 0.057	4.604 ± 0.717	1.131 ± 0.056	3.563 ± 0.357	40.976 ± 16.390	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.052 ± 0.022	-0.398 ± 0.073	-0.052 ± 0.018	-0.360 ± 0.051	-0.760 ± 0.048	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.023 ± 0.024	-0.340 ± 0.023	-0.023 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012	-0.023 ± 0.024	-0.340 ± 0.023	-0.023 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	3.023 ± 0.122	1.754 ± 0.107	2.963 ± 0.086	1.748 ± 0.114	0.457 ± 0.066	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	1.323 ± 0.153	1.437 ± 0.133	1.201 ± 0.130	1.317 ± 0.115	0.263 ± 0.078	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

J7 - L8

Metric	Base						ConvNeXt Atto										ConvNeXt Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.109 ± 0.022	0.736 ± 0.016	1.080 ± 0.022	0.715 ± 0.034	0.083 ± 0.005	0.935 ± 0.017	0.701 ± 0.040	0.938 ± 0.014	0.706 ± 0.030	0.002 ± 0.001	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000	0.955 ± 0.019	0.471 ± 0.051	0.955 ± 0.027	0.519 ± 0.036	0.000 ± 0.000
MAE	0.788 ± 0.023	0.565 ± 0.027	0.549 ± 0.025	0.543 ± 0.020	0.504 ± 0.024	0.561 ± 0.026	0.459 ± 0.026	0.477 ± 0.030	0.459 ± 0.025	0.456 ± 0.026	0.477 ± 0.033	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029	0.443 ± 0.022	0.499 ± 0.028	0.445 ± 0.022	0.469 ± 0.030	0.495 ± 0.029
RMSE	0.896 ± 0.022	0.711 ± 0.028	0.694 ± 0.033	0.682 ± 0.020	0.633 ± 0.034	0.724 ± 0.036	0.577 ± 0.036	0.601 ± 0.040	0.573 ± 0.035	0.575 ± 0.038	0.611 ± 0.040	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032	0.558 ± 0.032	0.635 ± 0.027	0.559 ± 0.032	0.597 ± 0.028	0.635 ± 0.032
NLL	1.541 ± 0.030	1.217 ± 0.086	2.658 ± 0.347	1.180 ± 0.097	2.275 ± 0.176	136.141 ± 88.256	1.000 ± 0.067	1.506 ± 0.236	0.982 ± 0.066	1.259 ± 0.123	16.035 ± 2.525	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174	0.979 ± 0.072	2.424 ± 0.300	0.979 ± 0.071	1.772 ± 0.147	86.117 ± 15.174
QCE 95	0.012 ± 0.017	-0.062 ± 0.009	-0.266 ± 0.031	-0.048 ± 0.006	-0.237 ± 0.026	-0.828 ± 0.025	-0.055 ± 0.016	-0.178 ± 0.042	-0.041 ± 0.014	-0.143 ± 0.031	-0.662 ± 0.034	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012	-0.025 ± 0.024	-0.340 ± 0.023	-0.025 ± 0.022	-0.265 ± 0.022	-0.808 ± 0.012
PI 95 μ	3.920 ± 0.000	3.094 ± 0.076	2.300 ± 0.124	3.005 ± 0.072	2.233 ± 0.134	0.285 ± 0.072	2.475 ± 0.060	1.998 ± 0.086	2.482 ± 0.062	1.977 ± 0.075	0.559 ± 0.026	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013	2.413 ± 0.081	1.411 ± 0.100	2.426 ± 0.080	1.524 ± 0.076	0.260 ± 0.013
PI 95 σ	0.000 ± 0.000	1.262 ± 0.107	1.615 ± 0.131	1.038 ± 0.139	1.473 ± 0.115	0.199 ± 0.084	0.569 ± 0.150	0.730 ± 0.061	0.505 ± 0.090	0.590 ± 0.032	0.192 ± 0.019	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009	0.317 ± 0.040	0.446 ± 0.030	0.338 ± 0.043	0.415 ± 0.026	0.090 ± 0.009

Poverty - 1000 Training Points

Distribution Shift - Rotation Invariant Scattering vs DINOv2

J5 - L8

Metric	Base						DINOv2 Small						DINOv2 Base					
	Scatterings						DINOv2 Small						DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		
LOSS	0.000 ± 0.000	1.061 ± 0.010	0.525 ± 0.024	1.045 ± 0.010	0.547 ± 0.040	0.029 ± 0.010	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001		
MAE	0.788 ± 0.023	0.552 ± 0.019	0.544 ± 0.050	0.528 ± 0.020	0.508 ± 0.051	0.558 ± 0.039	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019		
RMSE	0.896 ± 0.022	0.676 ± 0.022	0.700 ± 0.070	0.648 ± 0.025	0.653 ± 0.074	0.714 ± 0.059	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028		
NLL	1.541 ± 0.030	1.101 ± 0.056	3.897 ± 0.453	1.064 ± 0.045	3.054 ± 0.438	41.557 ± 25.712	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879		
QCE 95	0.012 ± 0.017	-0.038 ± 0.013	-0.363 ± 0.049	-0.038 ± 0.014	-0.335 ± 0.048	-0.753 ± 0.041	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060		
PI 95 μ	3.920 ± 0.000	2.881 ± 0.048	1.844 ± 0.118	2.801 ± 0.023	1.879 ± 0.110	0.505 ± 0.123	2.322 ± 0.049	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035		
PI 95 σ	0.000 ± 0.000	0.818 ± 0.101	1.440 ± 0.135	0.676 ± 0.036	1.419 ± 0.075	0.282 ± 0.034	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005		

J6 - L8

Metric	Base						Scattering						DINOv2 Small						DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin			
LOSS	0.000 ± 0.000	1.029 ± 0.012	0.707 ± 0.038	1.016 ± 0.019	0.697 ± 0.038	0.104 ± 0.004	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001	0.901 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019			
MAE	0.788 ± 0.023	0.532 ± 0.023	0.517 ± 0.034	0.518 ± 0.018	0.488 ± 0.033	0.564 ± 0.044	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019			
RMSE	0.896 ± 0.022	0.656 ± 0.031	0.656 ± 0.050	0.639 ± 0.026	0.612 ± 0.044	0.748 ± 0.081	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028			
NLL	1.541 ± 0.030	1.070 ± 0.051	2.477 ± 0.496	1.059 ± 0.049	2.114 ± 0.259	214.390 ± 121.202	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879			
QCE 95	0.012 ± 0.017	-0.032 ± 0.015	-0.247 ± 0.022	-0.031 ± 0.014	-0.236 ± 0.043	-0.864 ± 0.076	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060			
PI 95 μ	3.920 ± 0.000	2.743 ± 0.030	2.213 ± 0.153	2.734 ± 0.029	2.189 ± 0.141	0.223 ± 0.063	2.322 ± 0.049	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035			
PI 95 σ	0.000 ± 0.000	0.664 ± 0.032	1.456 ± 0.080	0.669 ± 0.038	1.397 ± 0.062	0.196 ± 0.101	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005			

J7 - L8

Metric	Base						DINOv2 Small						DINOv2 Base					
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin		
LOSS	0.000 ± 0.000	1.046 ± 0.025	0.883 ± 0.024	1.035 ± 0.021	0.850 ± 0.036	0.281 ± 0.009	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001		
MAE	0.788 ± 0.023	0.532 ± 0.019	0.535 ± 0.014	0.519 ± 0.023	0.490 ± 0.010	0.529 ± 0.033	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019		
RMSE	0.896 ± 0.022	0.659 ± 0.029	0.666 ± 0.037	0.642 ± 0.022	0.611 ± 0.009	0.701 ± 0.087	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028		
NLL	1.541 ± 0.030	1.111 ± 0.057	1.479 ± 0.181	1.086 ± 0.089	1.323 ± 0.122	1327.534 ± 819.659	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879		
QCE 95	0.012 ± 0.017	-0.035 ± 0.012	-0.148 ± 0.042	-0.029 ± 0.010	-0.128 ± 0.043	-0.908 ± 0.010	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060		
PI 95 μ	3.920 ± 0.000	2.826 ± 0.149	2.438 ± 0.150	2.770 ± 0.089	2.343 ± 0.160	0.089 ± 0.020	2.322 ± 0.049	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035		
PI 95 σ	0.000 ± 0.000	0.726 ± 0.219	1.164 ± 0.084	0.655 ± 0.112	1.062 ± 0.065	0.097 ± 0.060	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005		

Poverty - 1000 Training Points

Distribution Shift - Rotation Covariant Scattering vs DINOv2

J5 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.077 ± 0.016	0.501 ± 0.042	1.059 ± 0.013	0.401 ± 0.045	0.008 ± 0.003	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001
MAE	0.788 ± 0.023	0.551 ± 0.032	0.500 ± 0.049	0.530 ± 0.025	0.513 ± 0.042	0.511 ± 0.033	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019
RMSE	0.896 ± 0.022	0.690 ± 0.043	0.630 ± 0.051	0.658 ± 0.032	0.653 ± 0.048	0.646 ± 0.044	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028
NLL	1.541 ± 0.030	1.125 ± 0.043	3.021 ± 0.639	1.111 ± 0.037	3.367 ± 0.387	34.393 ± 0.812	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879
QCE 95	0.012 ± 0.017	-0.058 ± 0.016	-0.326 ± 0.044	-0.045 ± 0.020	-0.370 ± 0.042	-0.748 ± 0.026	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060
PI 95 μ	3.920 ± 0.000	2.930 ± 0.076	1.745 ± 0.146	2.950 ± 0.073	1.589 ± 0.056	0.419 ± 0.069	2.322 ± 0.040	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035
PI 95 σ	0.000 ± 0.000	1.080 ± 0.161	1.112 ± 0.056	1.079 ± 0.143	1.021 ± 0.055	0.139 ± 0.014	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005

J6 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.057 ± 0.003	0.465 ± 0.015	1.052 ± 0.006	0.476 ± 0.029	0.013 ± 0.005	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001
MAE	0.788 ± 0.023	0.550 ± 0.027	0.530 ± 0.056	0.531 ± 0.023	0.511 ± 0.044	0.521 ± 0.039	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019
RMSE	0.896 ± 0.022	0.687 ± 0.041	0.670 ± 0.064	0.660 ± 0.033	0.646 ± 0.051	0.657 ± 0.050	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028
NLL	1.541 ± 0.030	1.140 ± 0.057	4.604 ± 0.717	1.131 ± 0.056	3.563 ± 0.357	40.976 ± 16.390	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879
QCE 95	0.012 ± 0.017	-0.052 ± 0.022	-0.398 ± 0.073	-0.052 ± 0.018	-0.360 ± 0.051	-0.760 ± 0.048	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060
PI 95 μ	3.920 ± 0.000	3.023 ± 0.122	1.754 ± 0.107	2.963 ± 0.086	1.748 ± 0.114	0.457 ± 0.066	2.322 ± 0.049	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035
PI 95 σ	0.000 ± 0.000	1.323 ± 0.153	1.437 ± 0.133	1.201 ± 0.130	1.317 ± 0.115	0.263 ± 0.078	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005

J7 - L8

Metric	Base	Scattering					DINOv2 Small					DINOv2 Base				
	Base	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin	RBF	RBF ARD	Matern	Matern ARD	Ens Nonlin
LOSS	0.000 ± 0.000	1.109 ± 0.022	0.736 ± 0.016	1.080 ± 0.022	0.715 ± 0.034	0.083 ± 0.005	0.911 ± 0.013	0.676 ± 0.028	0.916 ± 0.013	0.702 ± 0.025	0.002 ± 0.001	0.906 ± 0.018	0.603 ± 0.021	0.911 ± 0.020	0.651 ± 0.021	0.004 ± 0.001
MAE	0.788 ± 0.023	0.565 ± 0.027	0.549 ± 0.025	0.543 ± 0.020	0.504 ± 0.024	0.561 ± 0.026	0.417 ± 0.021	0.451 ± 0.021	0.418 ± 0.020	0.427 ± 0.012	0.435 ± 0.028	0.419 ± 0.018	0.467 ± 0.017	0.422 ± 0.017	0.447 ± 0.020	0.438 ± 0.019
RMSE	0.896 ± 0.022	0.711 ± 0.028	0.694 ± 0.033	0.682 ± 0.020	0.633 ± 0.034	0.724 ± 0.036	0.532 ± 0.028	0.572 ± 0.026	0.531 ± 0.027	0.542 ± 0.018	0.558 ± 0.039	0.533 ± 0.020	0.595 ± 0.030	0.533 ± 0.020	0.568 ± 0.025	0.560 ± 0.028
NLL	1.541 ± 0.030	1.217 ± 0.086	2.658 ± 0.347	1.180 ± 0.097	2.275 ± 0.176	136.141 ± 88.256	0.929 ± 0.066	1.203 ± 0.140	0.923 ± 0.064	1.033 ± 0.110	11.157 ± 1.067	0.931 ± 0.048	1.503 ± 0.129	0.929 ± 0.050	1.202 ± 0.067	12.171 ± 1.879
QCE 95	0.012 ± 0.017	-0.062 ± 0.009	-0.266 ± 0.031	-0.048 ± 0.006	-0.237 ± 0.026	-0.828 ± 0.025	-0.034 ± 0.018	-0.144 ± 0.024	-0.025 ± 0.014	-0.099 ± 0.032	-0.602 ± 0.048	-0.032 ± 0.017	-0.200 ± 0.034	-0.024 ± 0.013	-0.153 ± 0.028	-0.635 ± 0.060
PI 95 μ	3.920 ± 0.000	3.094 ± 0.076	2.300 ± 0.124	3.005 ± 0.072	2.233 ± 0.134	0.285 ± 0.072	2.322 ± 0.049	1.833 ± 0.074	2.358 ± 0.045	1.904 ± 0.060	0.595 ± 0.035	2.315 ± 0.059	1.706 ± 0.080	2.347 ± 0.058	1.807 ± 0.069	0.564 ± 0.035
PI 95 σ	0.000 ± 0.000	1.262 ± 0.107	1.615 ± 0.131	1.038 ± 0.139	1.473 ± 0.115	0.199 ± 0.084	0.222 ± 0.030	0.397 ± 0.021	0.225 ± 0.029	0.308 ± 0.022	0.165 ± 0.007	0.215 ± 0.024	0.351 ± 0.028	0.217 ± 0.019	0.280 ± 0.023	0.167 ± 0.005